THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND TWO
CROSSNUMBER ROUND (ACROSS)

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ACROSS

1. The value of $f(0)$, given that $f(x) + \displaystyle\int_{-1}^{1} f(t)\,\mathrm{d}t = 24x^2 - 25x + 52$.

2. The value of ab, where $a, b \geq 1$ are coprime integers less than 20 such that

$$\int_{-\infty}^{\infty} \frac{1}{\left(|x| + \sqrt[b]{a}\right)\left(\ln\left(|x| + \sqrt[b]{a}\right)\right)^2}\,\mathrm{d}x = \frac{14}{\ln 3 + 2\ln 2}.$$

4. A perfect cube.

6. Shares a common factor (other than 1) with the smallest positive integer n for which

$$\int_0^{\infty} \cos(x) \prod_{k=1}^{n} \frac{\sin(x/(2k-1))}{x/(2k-1)}\,\mathrm{d}x \neq \frac{\pi}{4}.$$

7. $\sqrt{13\text{ DOWN}}$.

8. (7 ACROSS)a + (5 DOWN)b, where $\displaystyle\int_0^1 \int_0^1 (xy)^{xy}\,\mathrm{d}x\,\mathrm{d}y = \frac{a}{b}\int_0^1 x^x\,\mathrm{d}x$ with $\frac{a}{b}$ in simplest form.

11. $(b^a)^2$, where $\frac{a}{b}$ is the greatest positive number in simplest terms such that

$$\frac{\lfloor (2\text{ ACROSS})/10 \rfloor}{20}\int_0^1 f(x)\,\mathrm{d}x + \frac{2}{3}\int_0^{\frac{3}{5}} f(x)\,\mathrm{d}x \geq \int_0^{\frac{a}{b}} f(x)\,\mathrm{d}x$$

for all continuous convex $[0,1] \to \mathbb{R}$ functions f.

12. A 5-digit number $\overline{abcde}$ where $\overline{de} = a + b + c$.

In the answers to the following clues, the digits 1-9 appear once and exactly once.

15. A sexy prime (i.e. a prime number differing from another prime number by exactly 6).

18. An even number.

19. A perfect square greater than 20×25.